Lecture 9: NP-completeness, Reductions

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Recap: NP

The class **NP**

Definition: A verifier for a language A is a TM V, where

$$A = \{w \mid \exists C \text{ s.t. } V \text{ accepts } \langle w, C \rangle\}.$$

A polynomial time verifier runs in polynomial time in $|w|$.

Definition: **NP** is the class of languages with polynomial time verifiers

Non-deterministic Turing Machines

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An NTM is a **polynomial time** NTM if the running time its longest computation on x is **polynomial in $|\mathbf{x}|$**.

Nondeterministic deciders $\iff$ Verifiers

Theorem: For any language $L \subseteq \Sigma^*$,

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V accepts $\langle x, C \rangle$ iff C is the accepting path of N on x.

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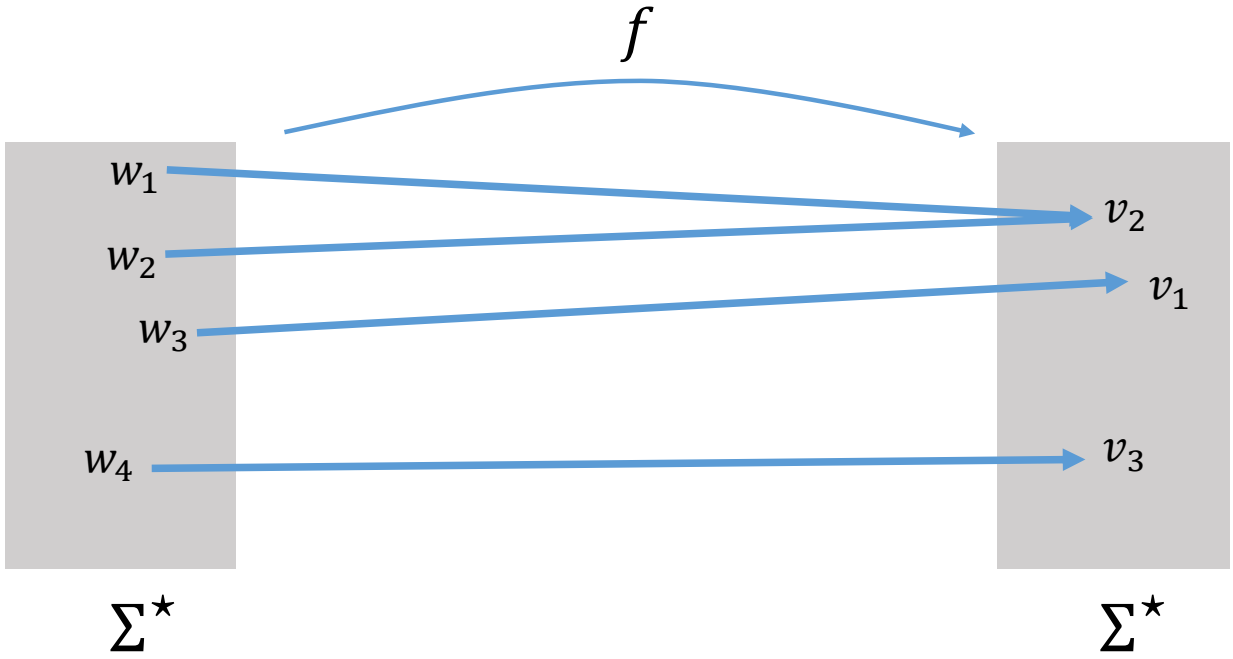
$\therefore$ SAT, GI, INDSET are all in NP.

NP-completeness

Poly-time Reductions (1): Poly-time Computability



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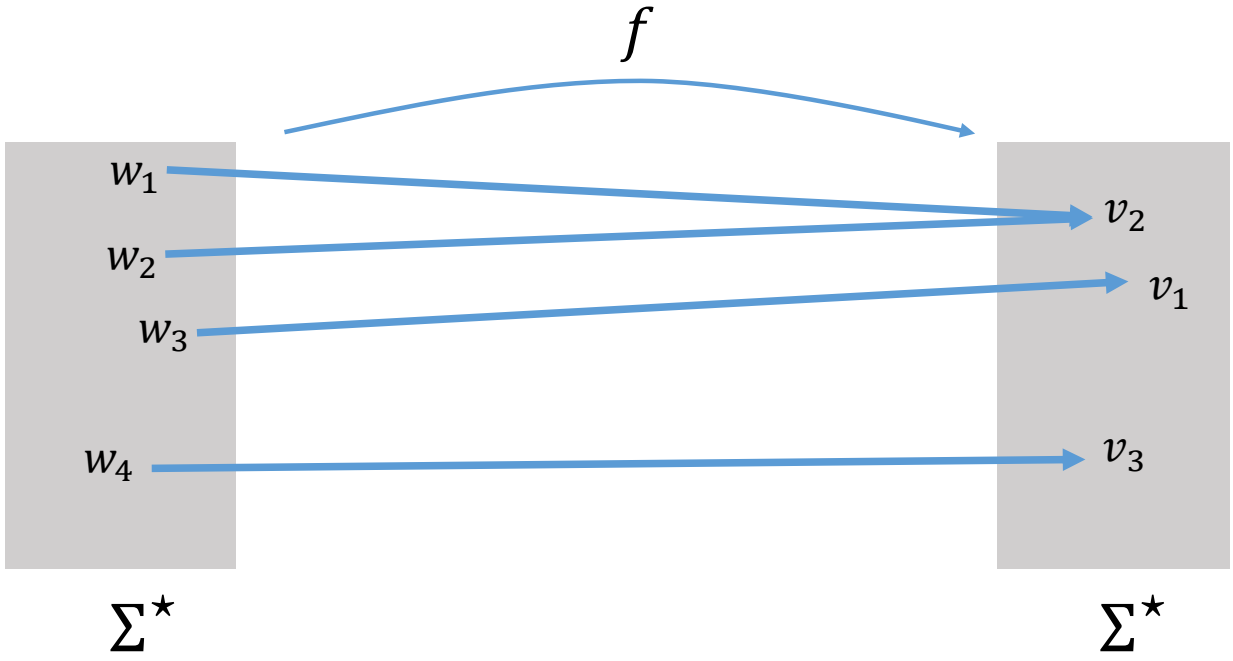
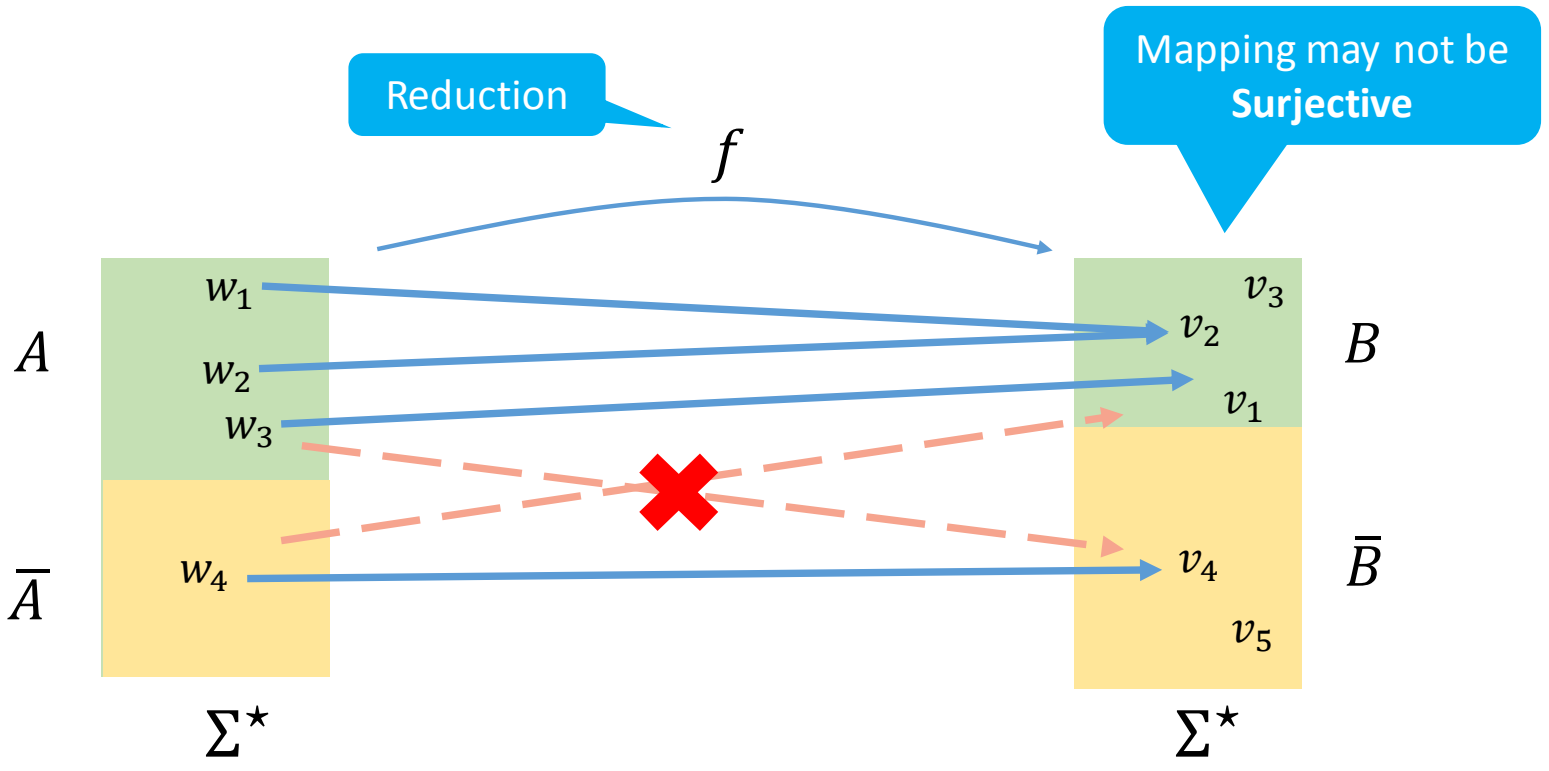


Definition: A function $f : \Sigma^* \to \Sigma^*$ is a *polynomial time computable function* if some poly-time TM M, on every input w halts with just $f(w)$ on its tape.

Poly-time Reductions (2): Correctness



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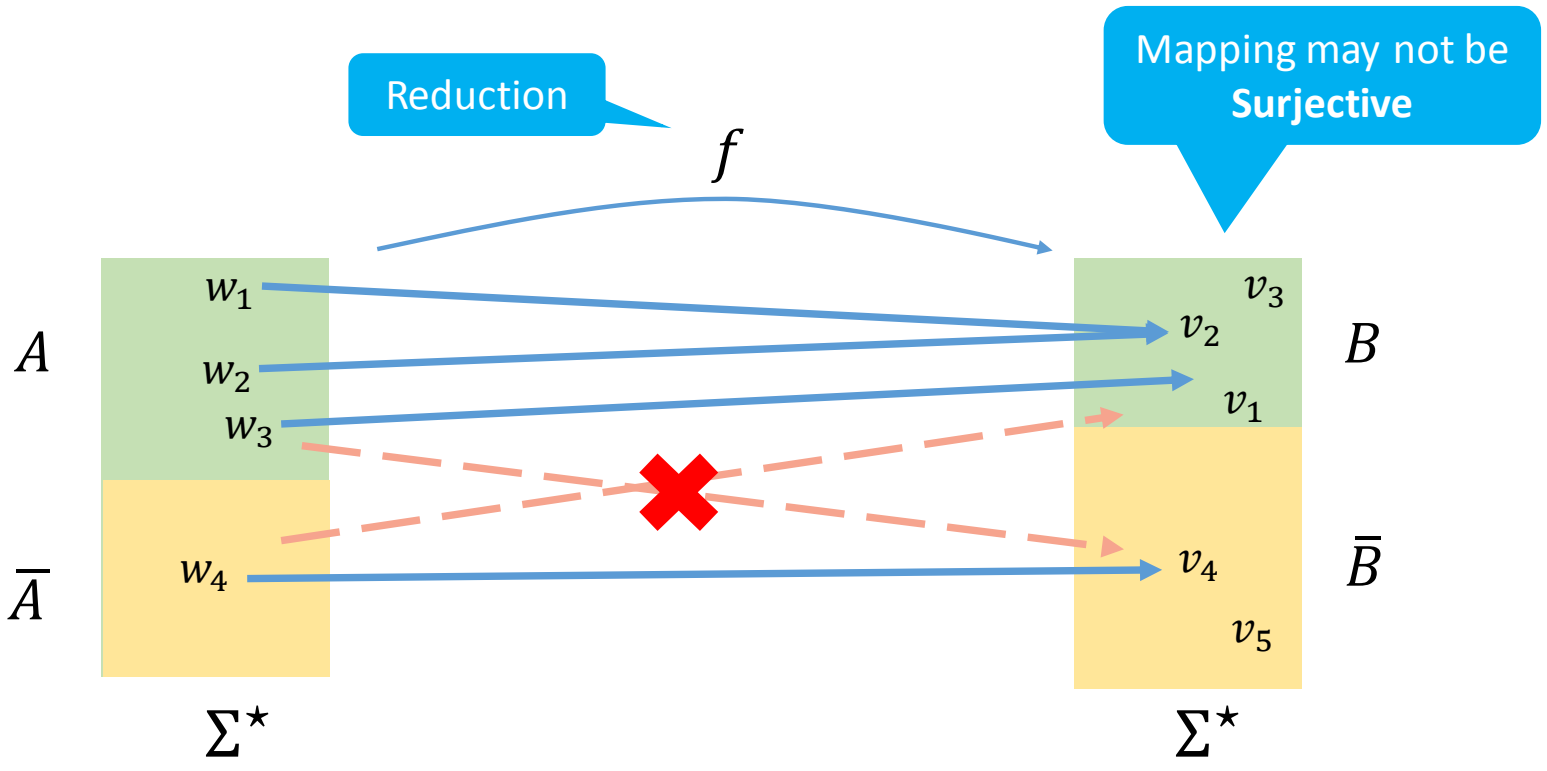

Definition: Language A is **poly-time mapping reducible** to language B, written $A \leq_P B$, if there is a poly-time computable function $f : \Sigma^* \to \Sigma^*$, such that for every $w \in \Sigma^*$:

$$w \in A \Leftrightarrow f(w) \in B$$

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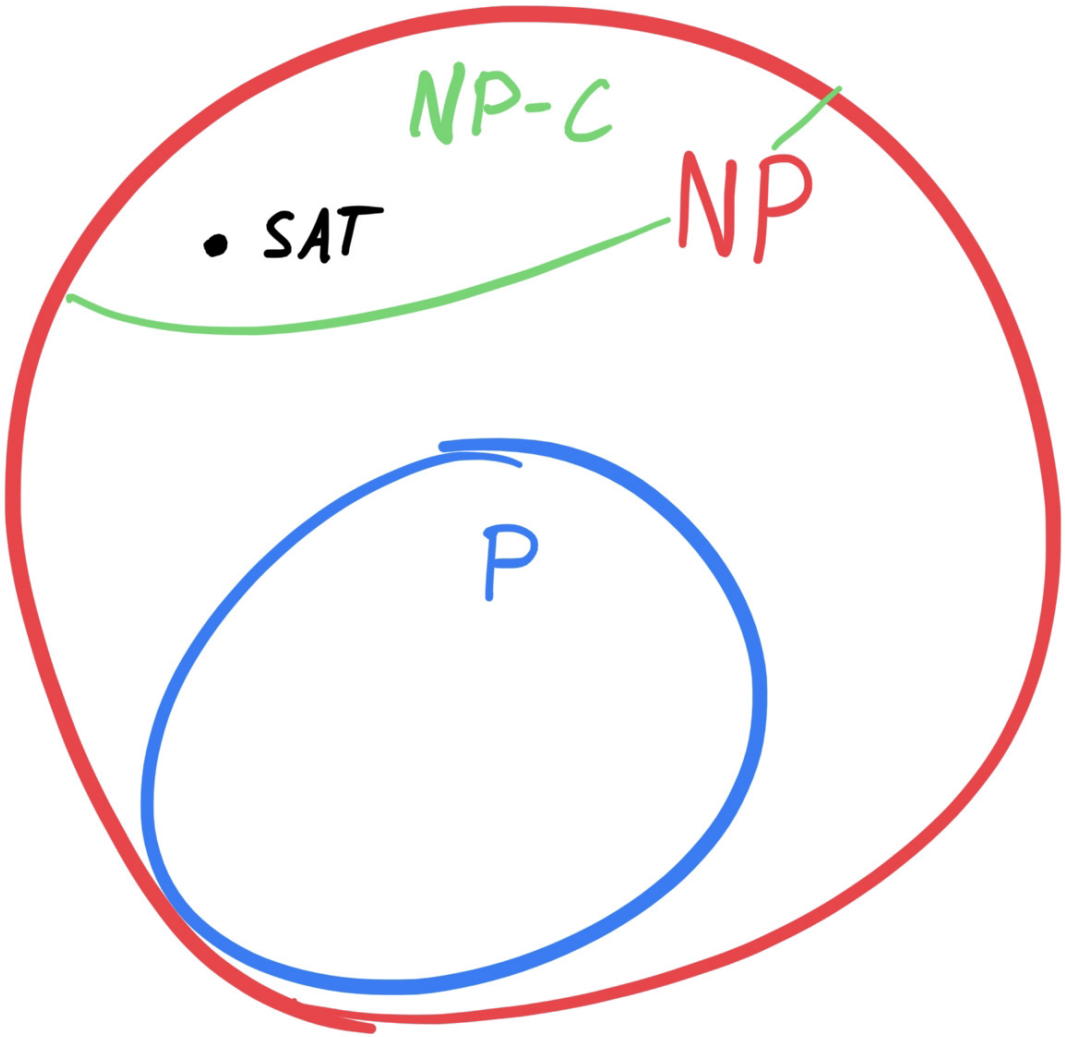
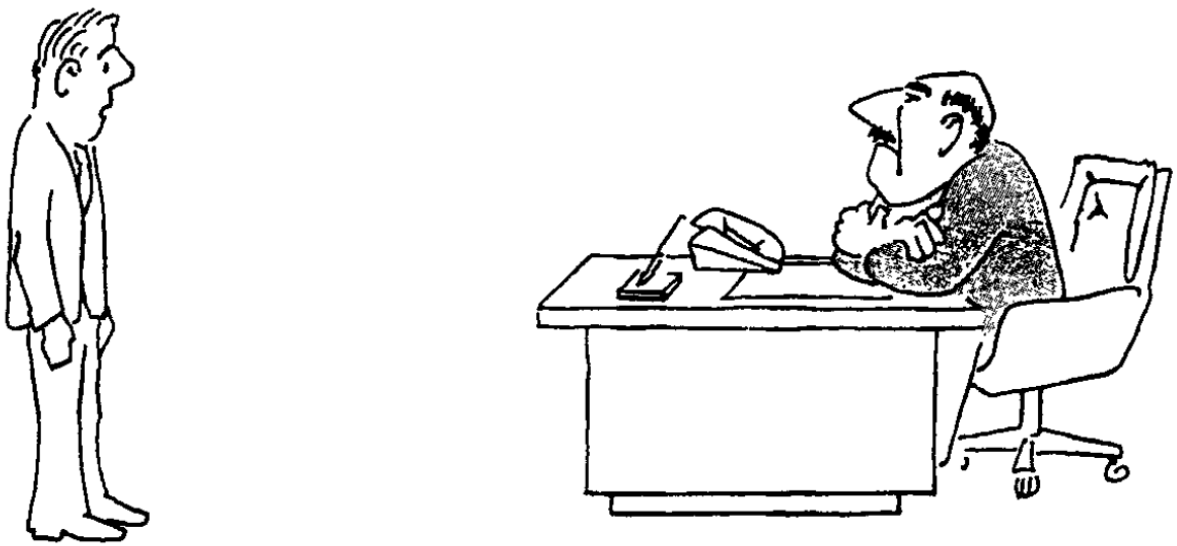
To show L is **NP**-complete:

- **[NP membership]** Give a poly-time verifier for L.
- **[NP hardness]** Show $C \leq_P L$ for some **NP**-complete language C (not the other way around)

Big picture

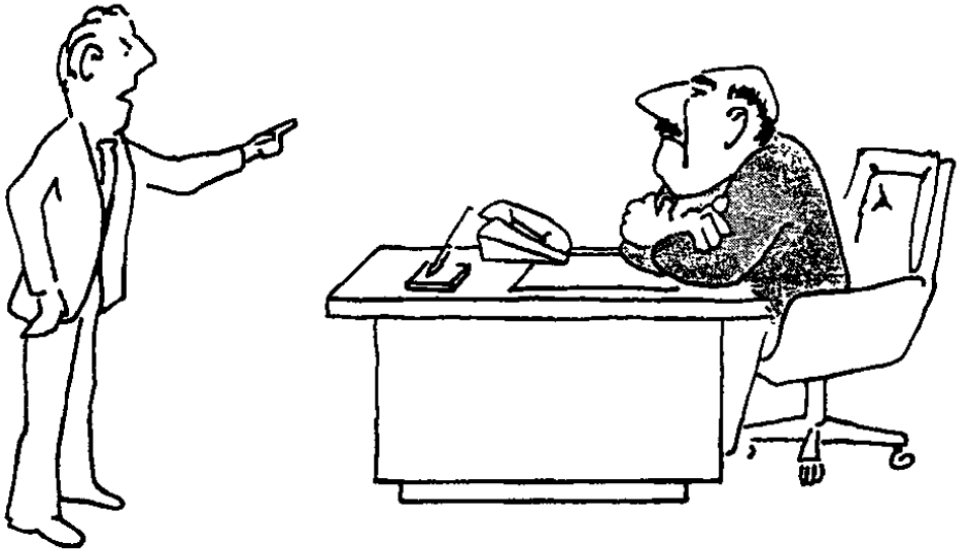


Takehome message



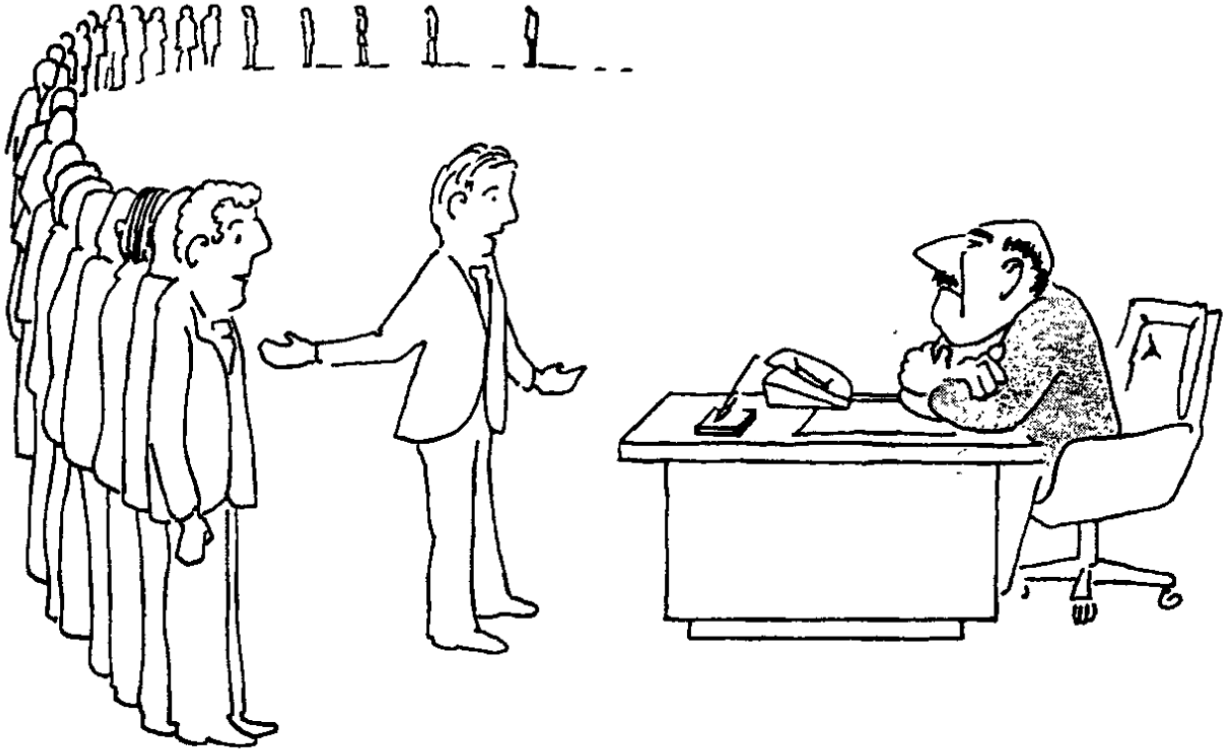
"I can't find an efficient algorithm, I guess I'm just too dumb."

Takehome message



"I can't find an efficient algorithm, because no such algorithm is possible!"

Takehome message



"I can't find an efficient algorithm, but neither can all these famous people."

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$$k\text{SAT} = \{\langle\varphi\rangle \ : \ \varphi \text{ is satisfiable and each clause of } \varphi \text{ contains } \leq k \text{ literals}\}$$

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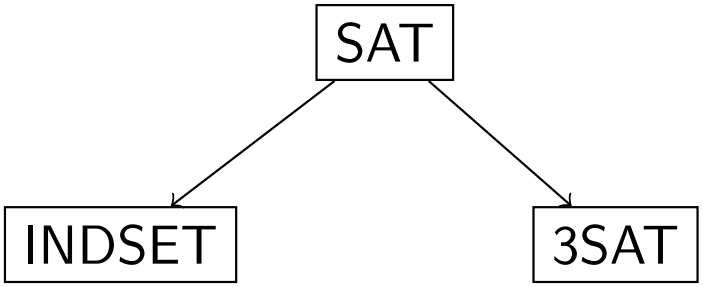
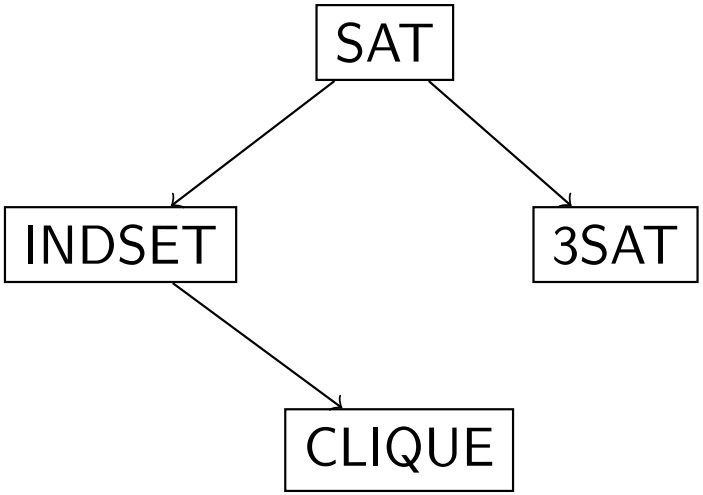
What is the runtime?

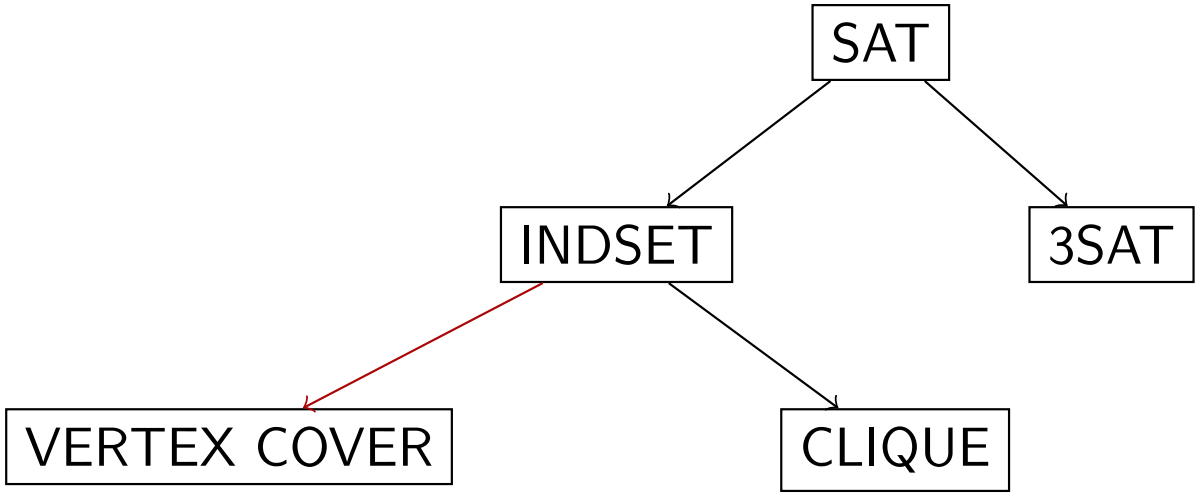
Does SAT $\leq_P$ 2SAT analogously?

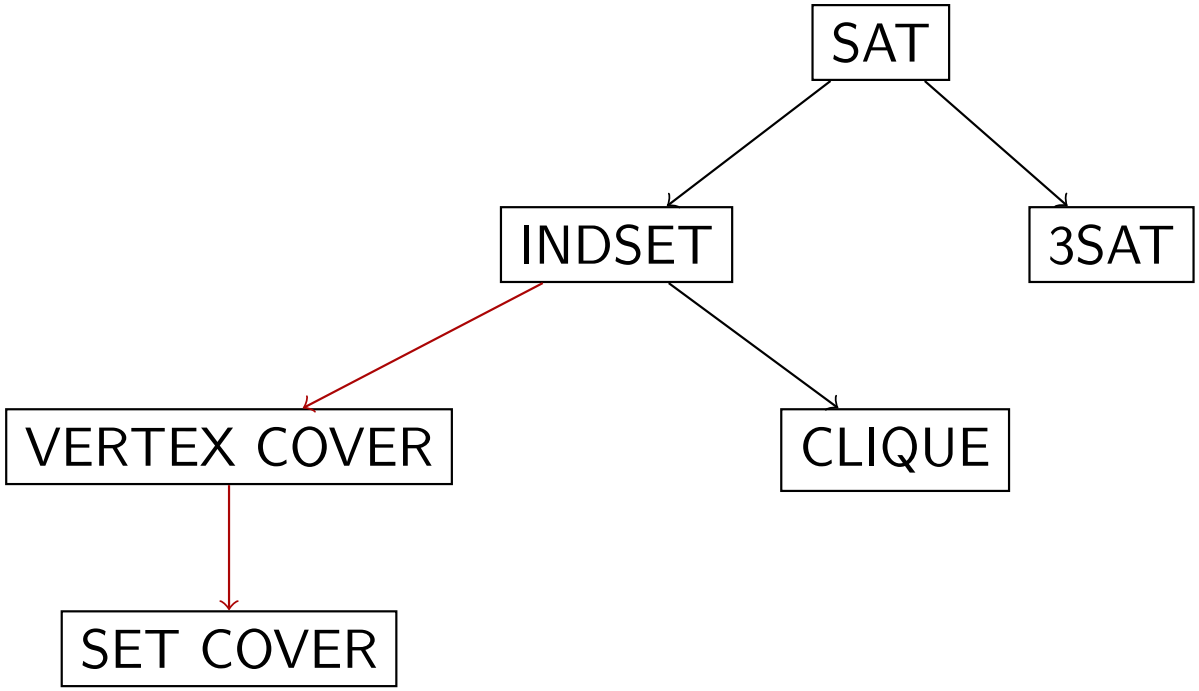
$\boxed{\text{SAT}}$

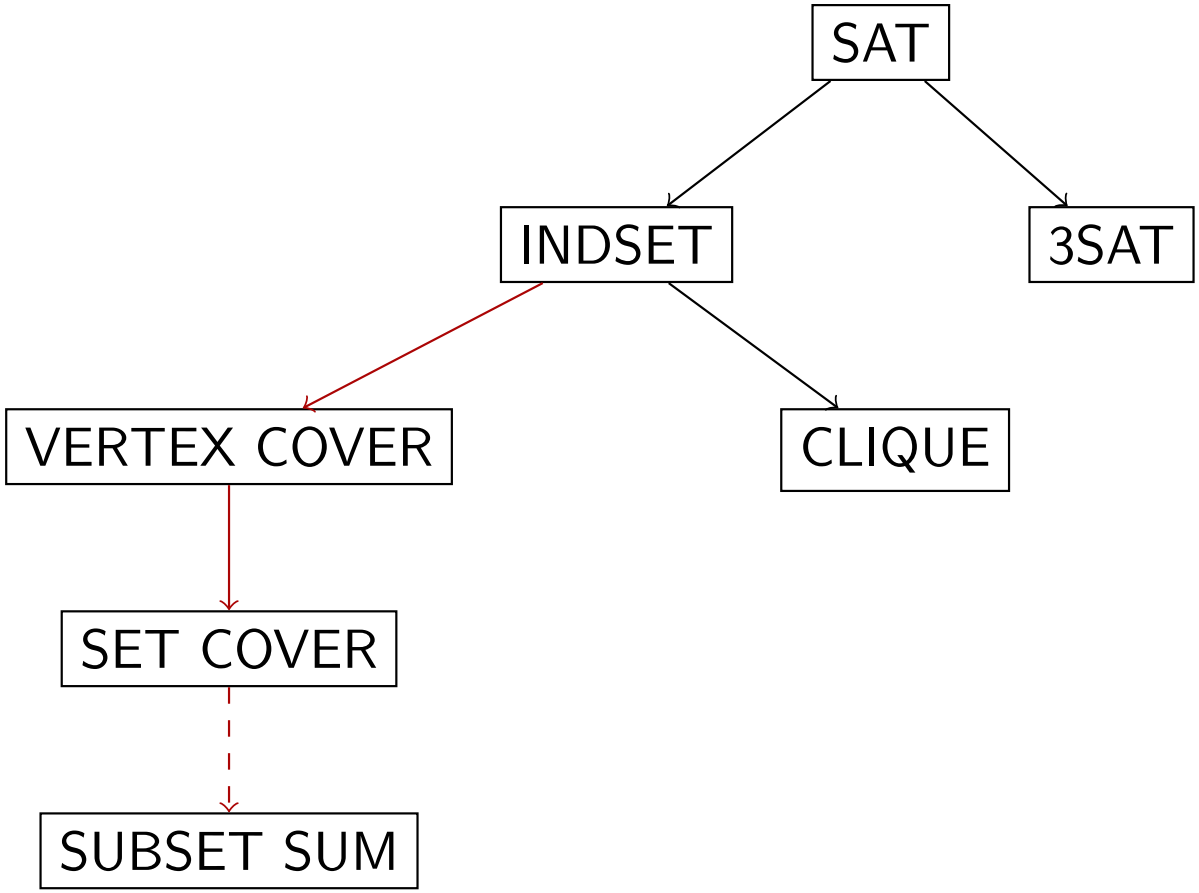
$$\boxed{\text{SAT}} \longrightarrow \boxed{\text{3SAT}}$$

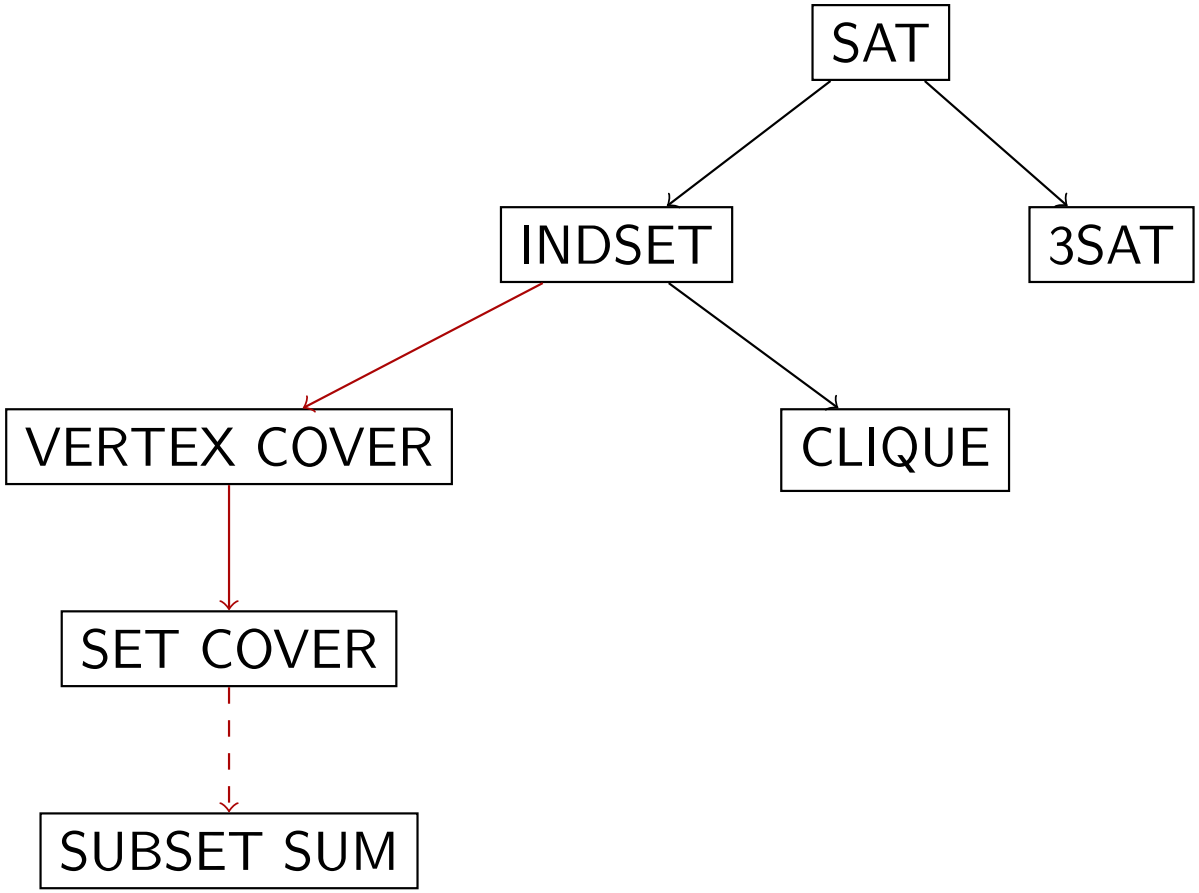

SAT
INDSET
3SAT


SAT
INDSET
3SAT
CLIQUE

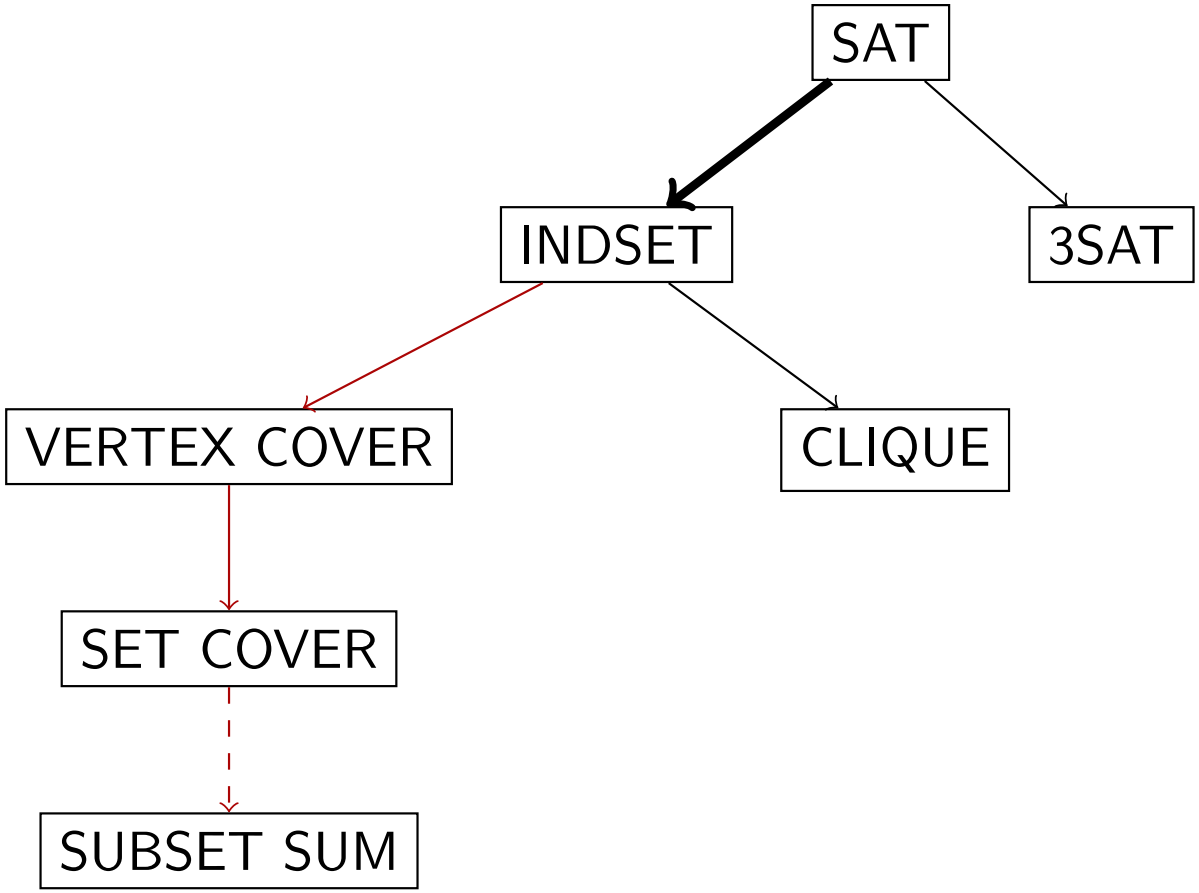

SAT
INDSET
3SAT
VERTEX COVER
CLIQUE


SAT
INDSET
3SAT
VERTEX COVER
CLIQUE
SET COVER


SAT
INDSET
3SAT
VERTEX COVER
CLIQUE
SET COVER
SUBSET SUM



and many more . . .


SAT
INDSET
3SAT
VERTEX COVER
CLIQUE
SET COVER
SUBSET SUM

INDSET is NP-complete

We have already seen that

$$\text{INDSET} = \{\langle G, k\rangle : G \text{ has an independet set of size } k\}$$

is in **NP** and so it remains to give a poly-time reduction from an **NP**-complete language

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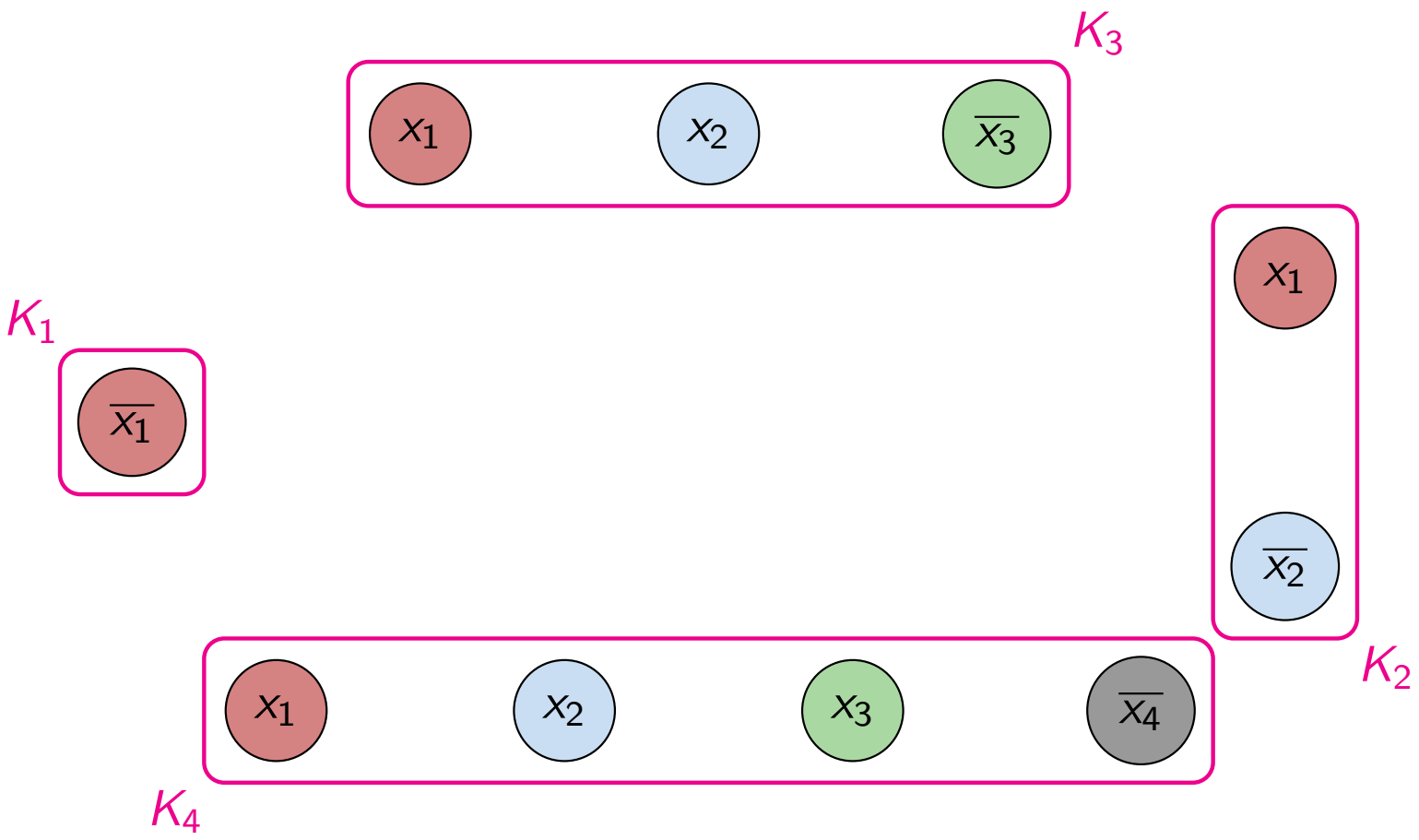

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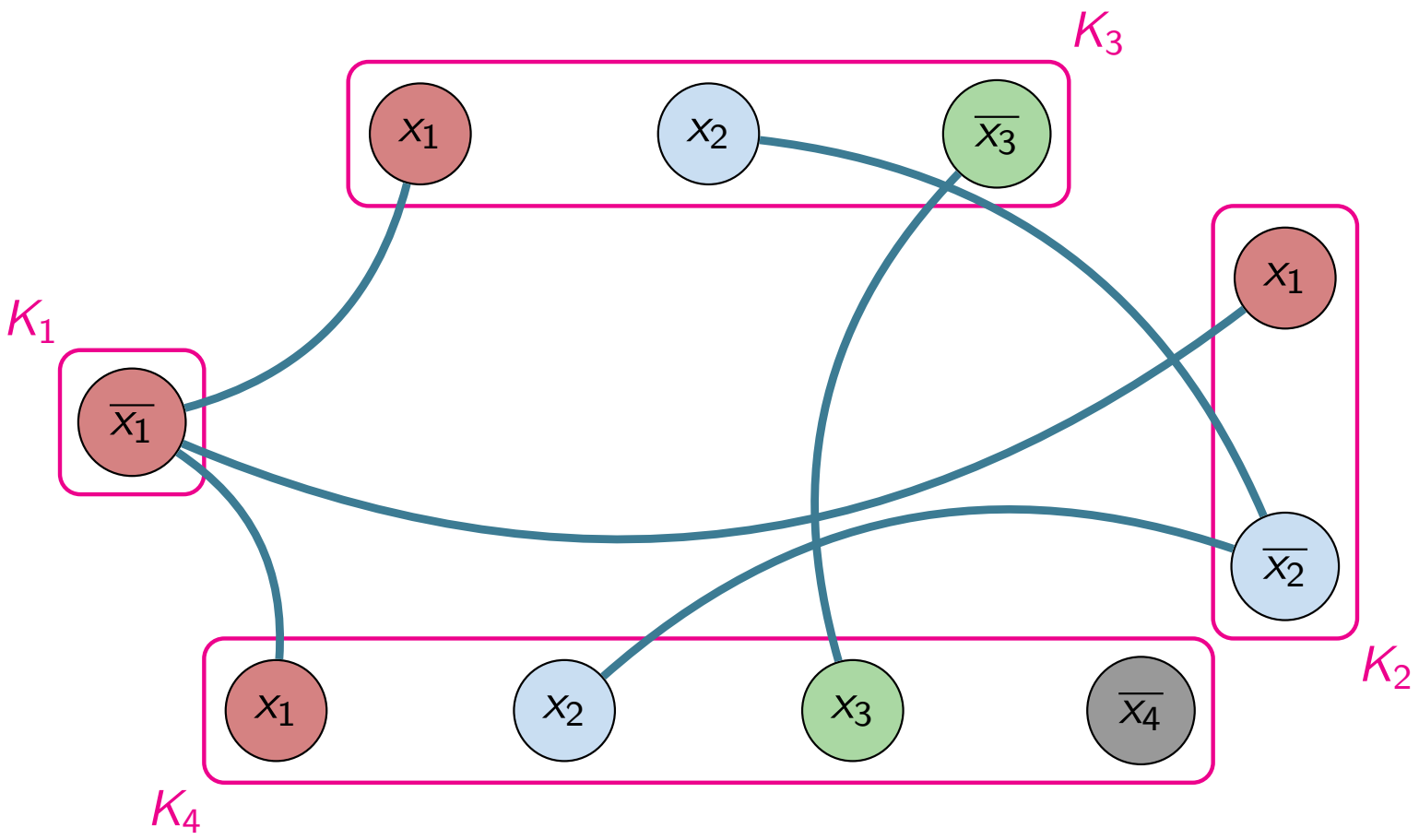

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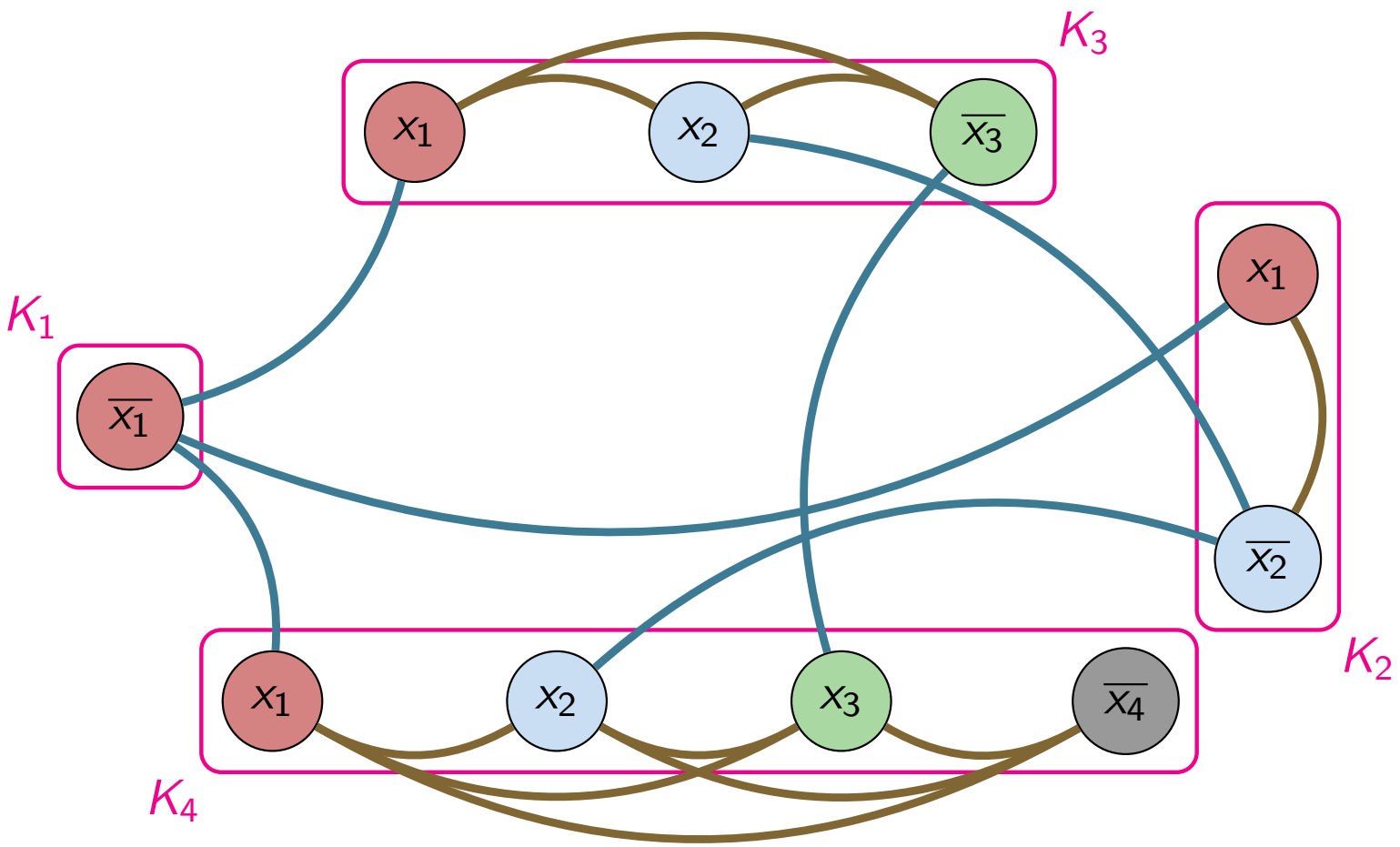

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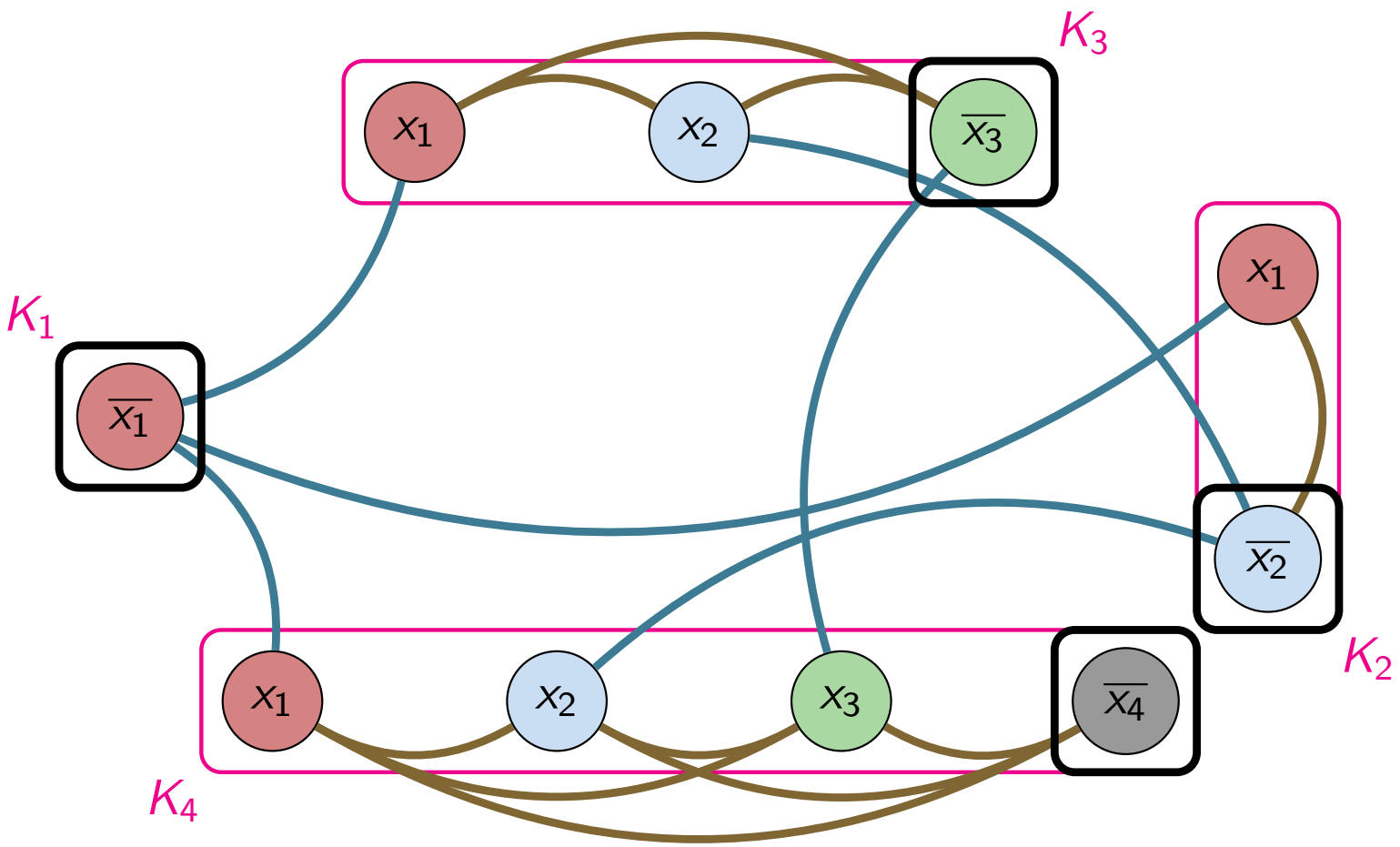

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Reduction f: On input φ,

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1. Let G be the graph generated as follows.

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INDSET is NP-complete

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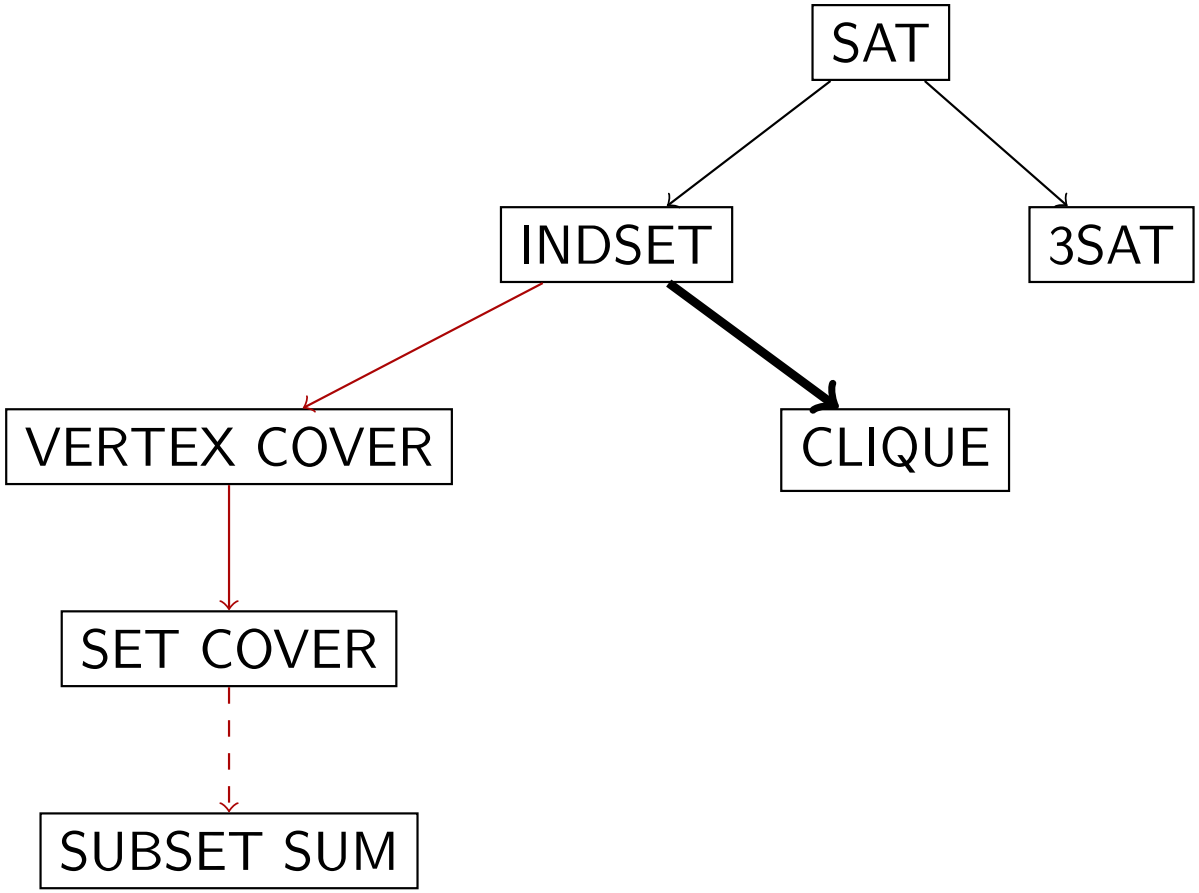
Proof: C: independent set in G, $|C| = m$. C contains one vertex from each group. Set the corresponding literals to `true` to get a satisfying assignment.


SAT
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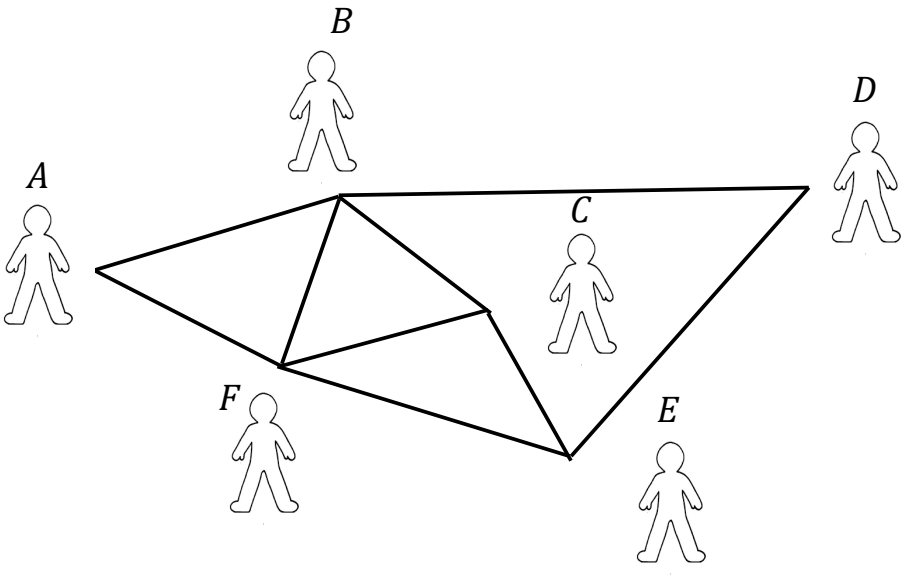
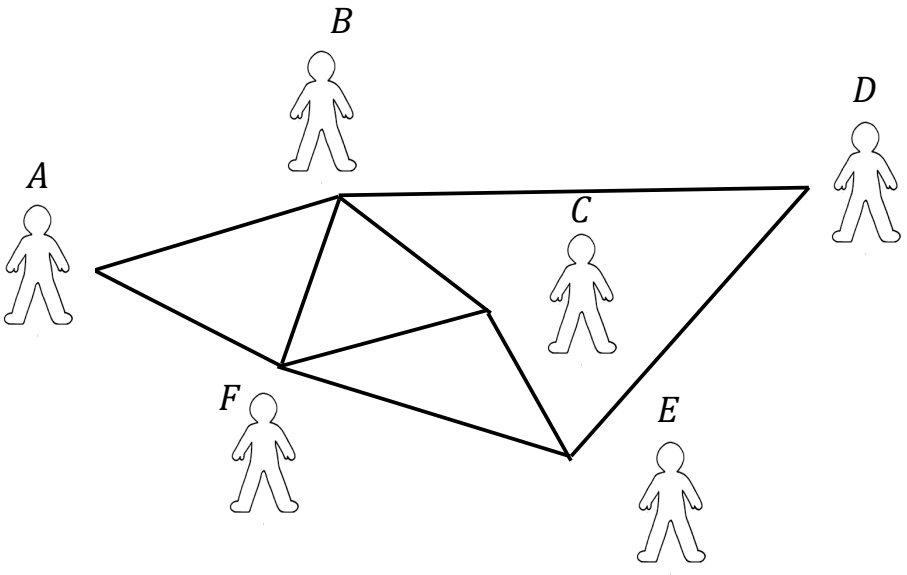
CLIQUE



CLIQUE

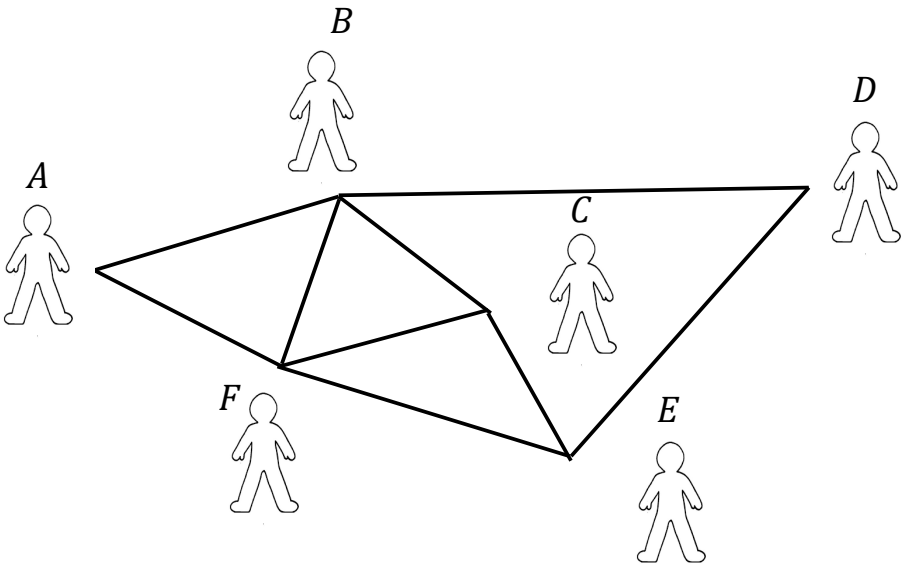


A,B,F is a 3-clique

Definition: A k-clique is a subset of k pairwise connected vertices

$$\text{CLIQUE} = \{\langle G, k\rangle \mid G \text{ has a clique of size } k\}$$

CLIQUE



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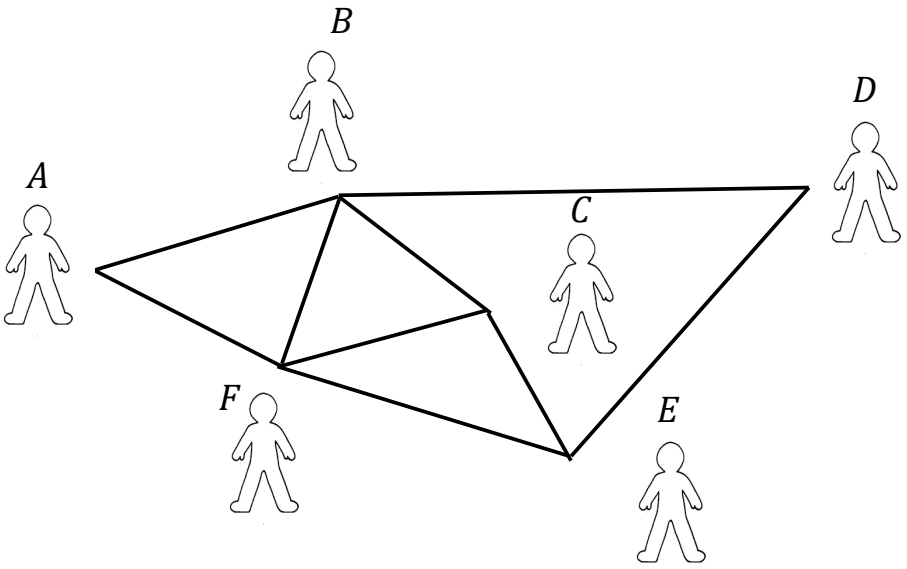
CLIQUE



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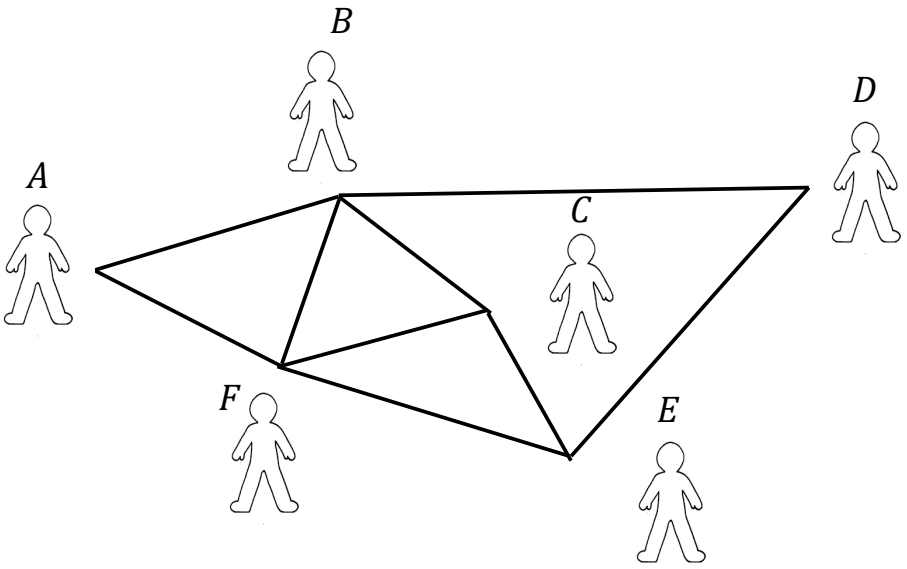
CLIQUE



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CLIQUE is **NP**-Complete

Theorem: INDSET $\leq_p$ CLIQUE

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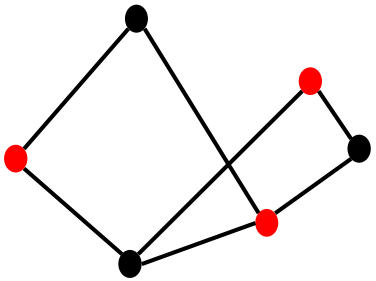
Goal find a poly-time reduction from INDSET to CLIQUE

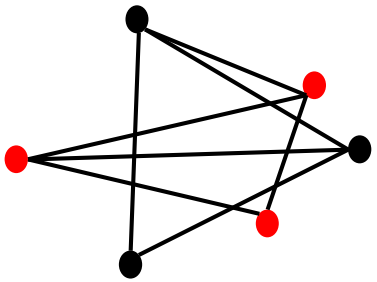
INDSET vs CLIQUE

For a graph G:

Independent set: A subset of pairwise **non-adjacent** vertices

Clique: A subset of pairwise **adjacent** vertices





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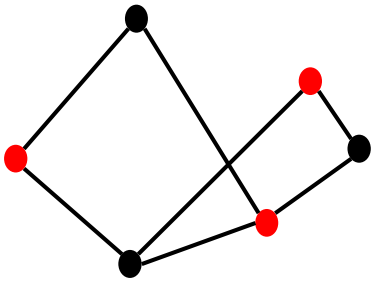

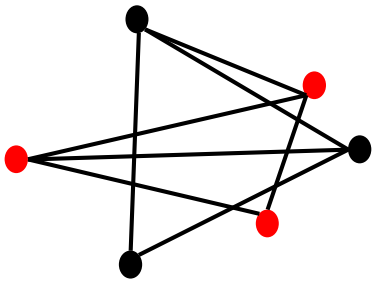

Def (Complement): The *complement* of a graph $G = (V, E)$ is a graph $\bar{G} = (V, \bar{E})$ with same vertex set and edge set $\bar{E}$ s.t. $uv \in \bar{E}$ iff $uv \notin E$

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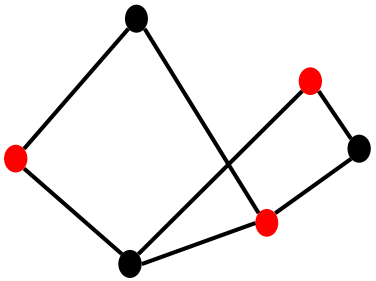

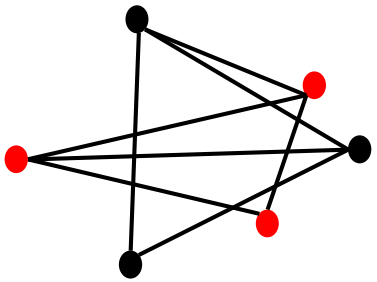

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Observation: If G is a graph and $\bar{G}$ its complement, then a subset S of the vertices of G is an independent set iff S is a clique of $\bar{G}$

INDSET $\leq_p$ CLIQUE

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Reduction: $f(\langle G = (V, E), k\rangle) := \langle \bar{G} = (V, \bar{E}), k\rangle$

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Correctness: $\langle G, k\rangle \in \text{INDSET} \Leftrightarrow \langle \bar{G}, k\rangle \in \text{CLIQUE}$

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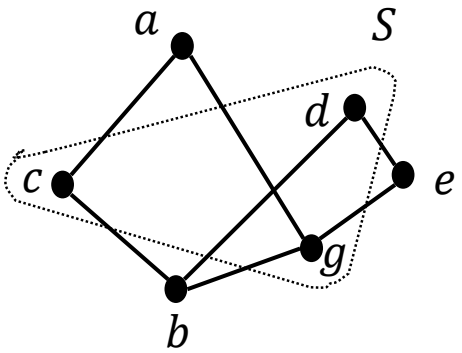
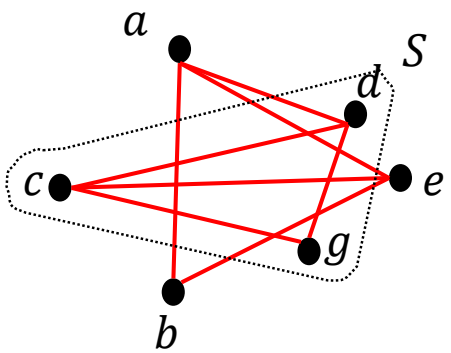
Proof of correctness:

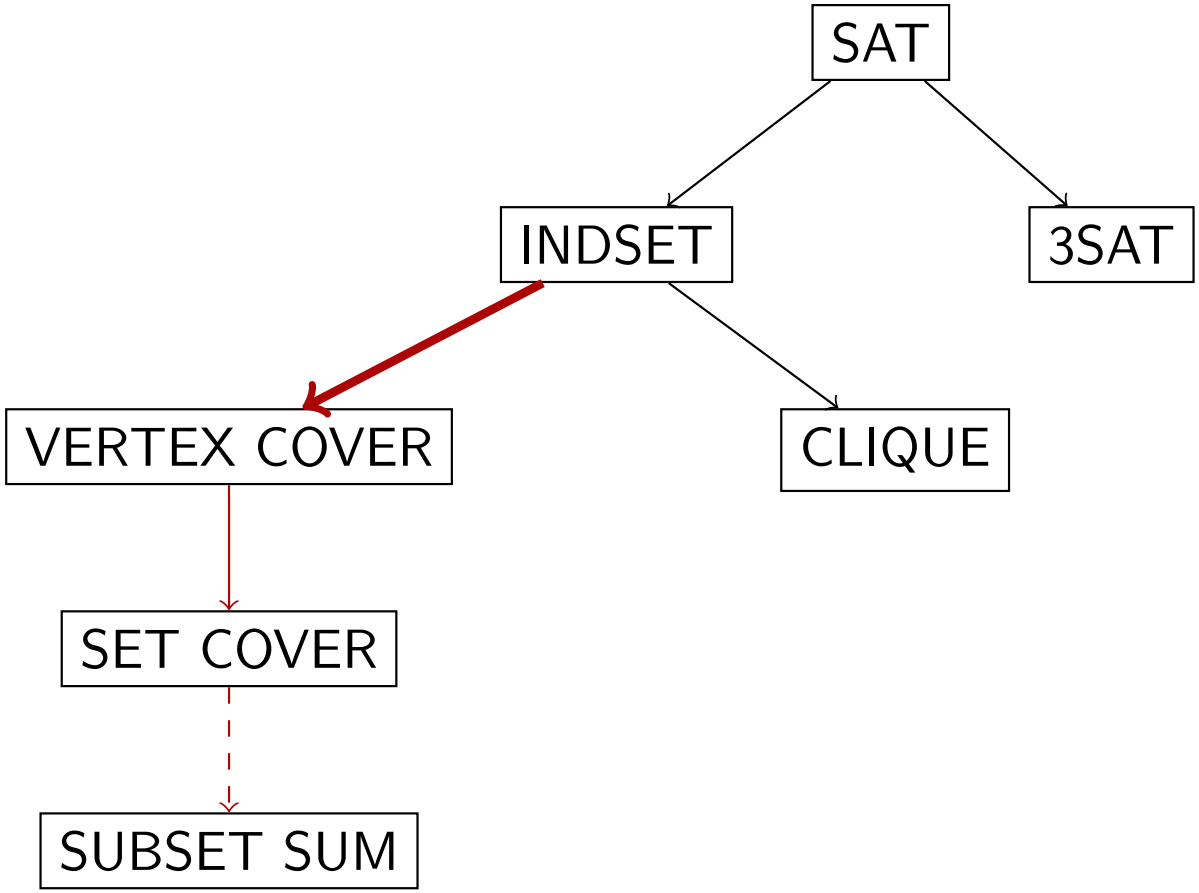


$\langle G, k\rangle \in INDSET, |S| = k$

$\Leftrightarrow$



$\langle \bar{G}, k\rangle \in CLIQUE$


SAT
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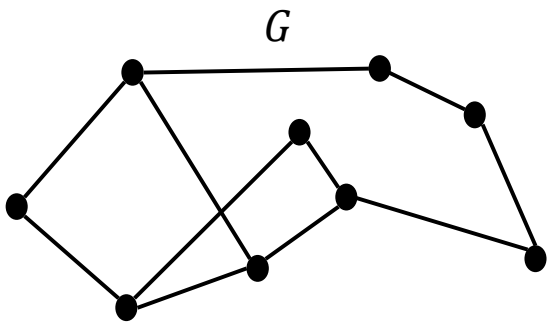
Vertex Cover

Definition: For a graph $G = (V, E)$, a *vertex cover* is a subset S of V such that every edge of G is incident to at least one vertex in S

Example



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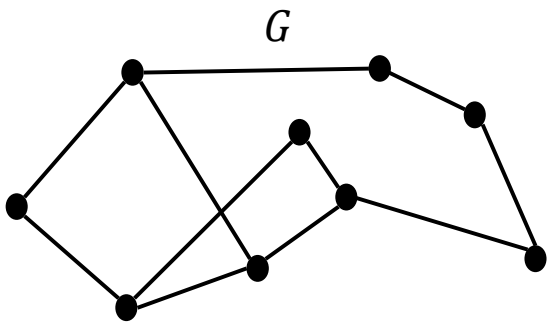

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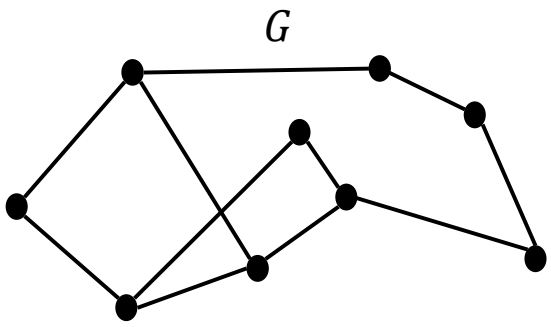

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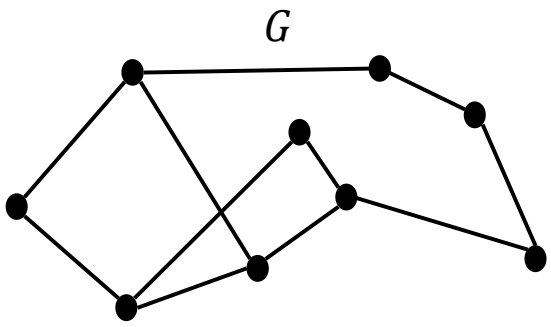

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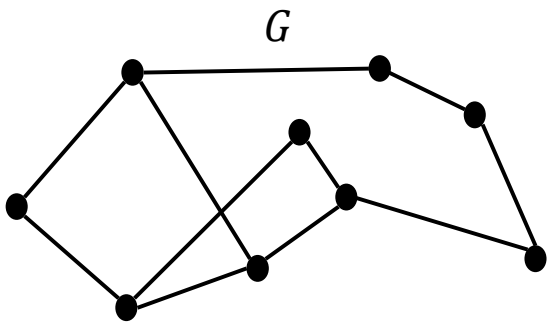

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VERTEX COVER is **NP**-Complete

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STEP 1: VERTEX COVER $\in$ **NP**

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STEP 2: INDSET $\leq_p$ VERTEX COVER

How to reduce INDSET to VERTEX COVER?

INDSET vs VERTEX COVER

INDSET vs VERTEX COVER

Lemma: For every graph G, a subset S of the vertices is a vertex cover if and only if $\bar{S}$ is an independent set where $\bar{S} = V \setminus S$

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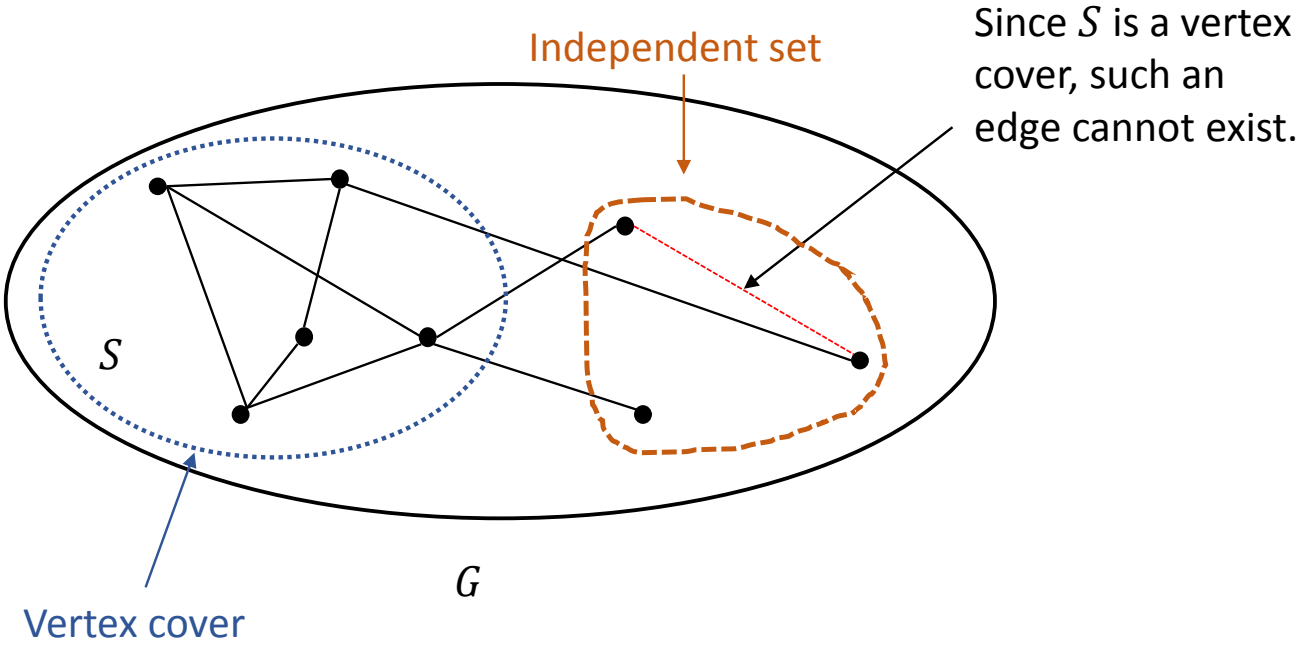

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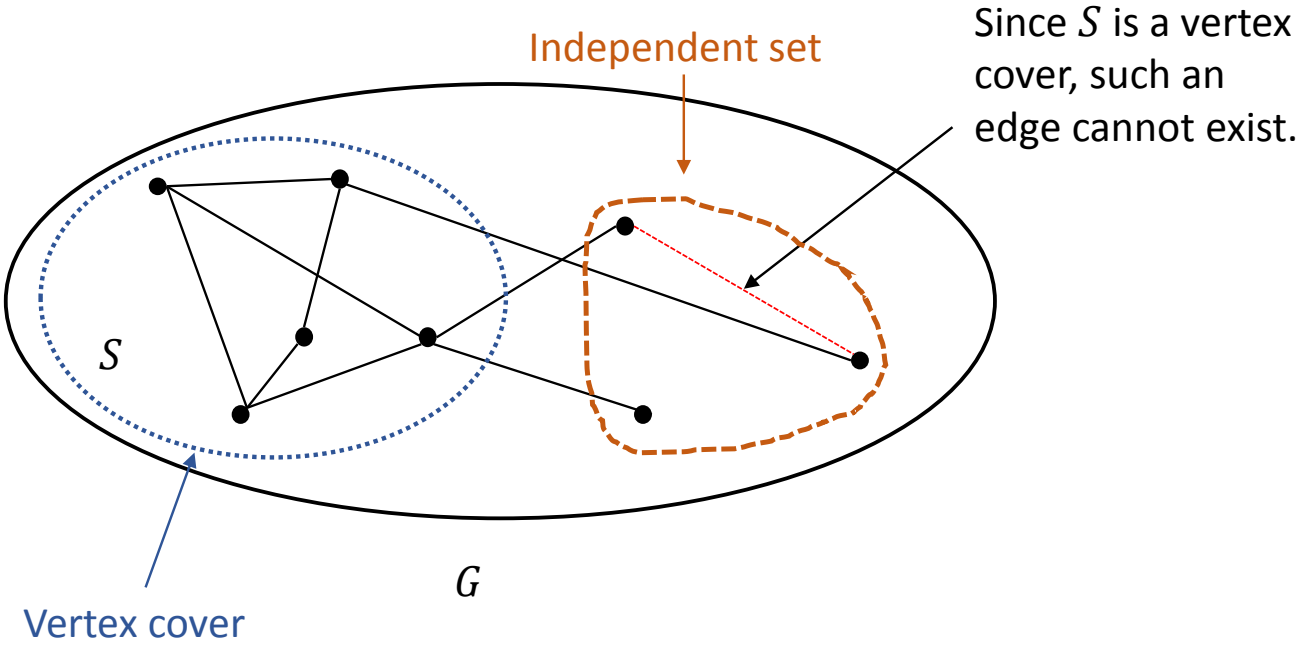

Corollary: For every graph G and a positive integer k, G has an independent set of size k iff G has a vertex cover of size $n - k$

INDSET $\leq_p$ VERTEX COVER

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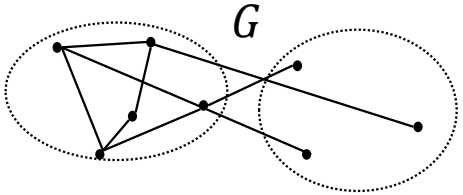
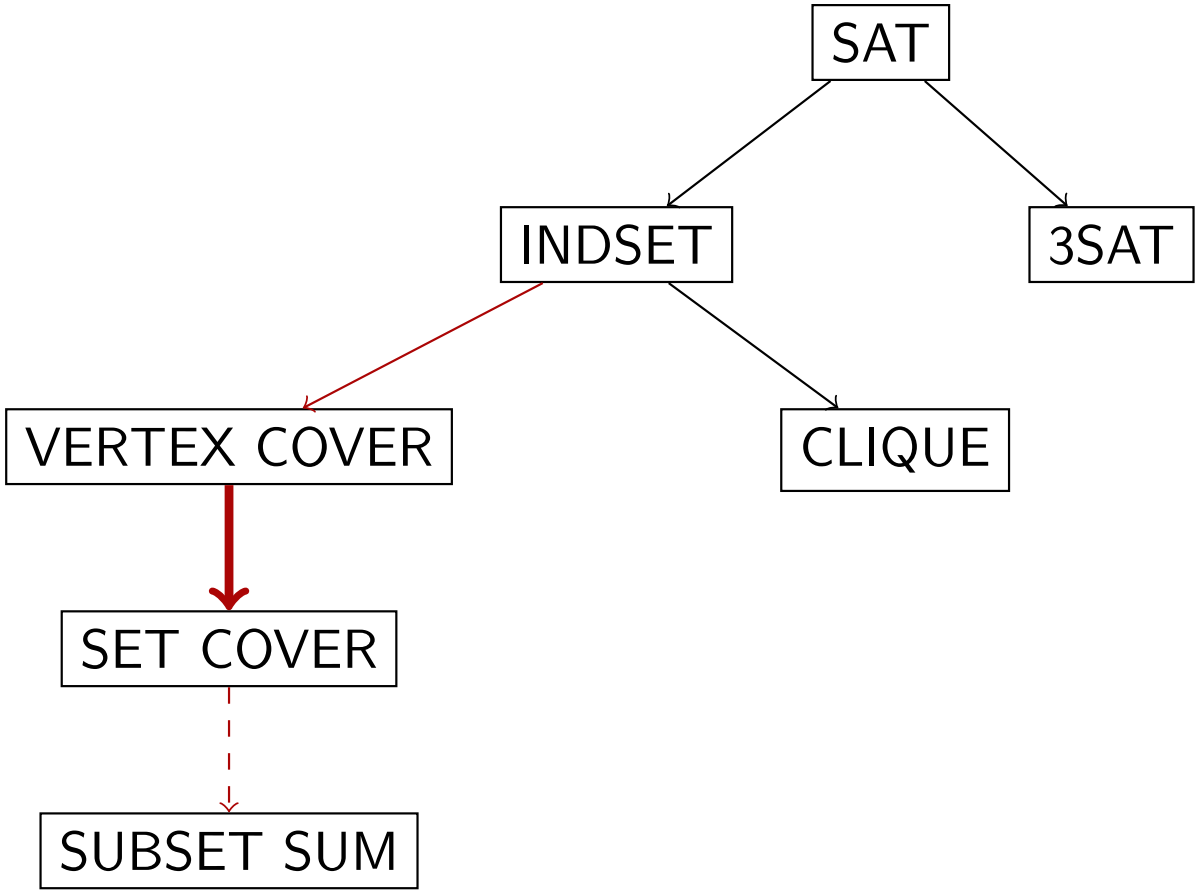
Correctness: Lemma/Corollary on previous slide!

$\langle G, k\rangle \in INDSET \iff$ 

$\iff \langle G, n-k\rangle \in VERTEX\ COVER$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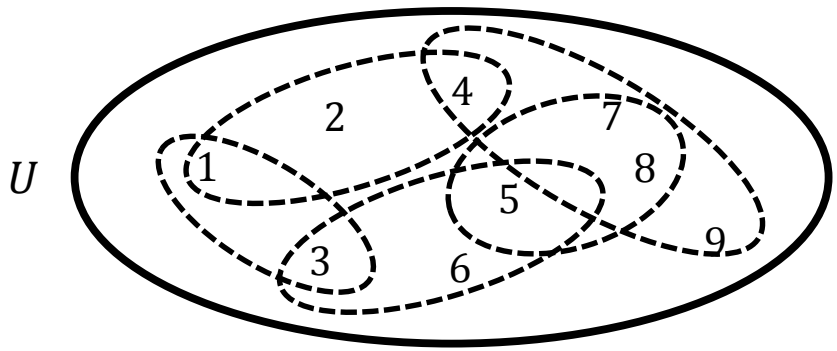
Set Cover

Definition: Let $U = \{1, \ldots, n\}$ and $\mathcal{F} = \{T_1, \ldots, T_m\}$ be a family of subsets $\forall i, T_i \subseteq U$

A subset $\{T_{i_1}, \ldots, T_{i_k}\} \subseteq \mathcal{F}$ is called a set cover of size k if $\bigcup_{j=1}^{k} T_{i_j} = U$



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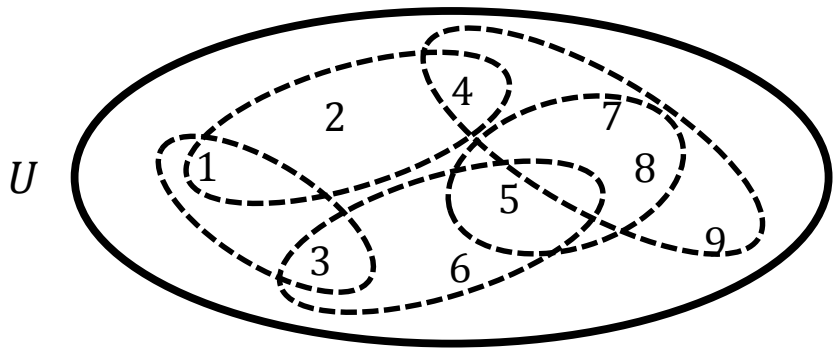

$$\text{SET COVER} = \{\langle U, \mathcal{F}, k\rangle \mid \mathcal{F} \text{ contains a set cover of size } k\}$$

Example:

- $\langle U, \mathcal{F}, 3\rangle \in$ SET COVER?

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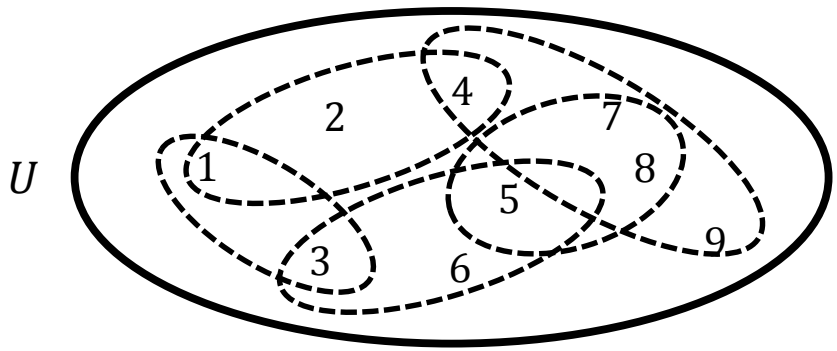

$$\text{SET COVER} = \{\langle U, \mathcal{F}, k\rangle \mid \mathcal{F} \text{ contains a set cover of size } k\}$$

Example:

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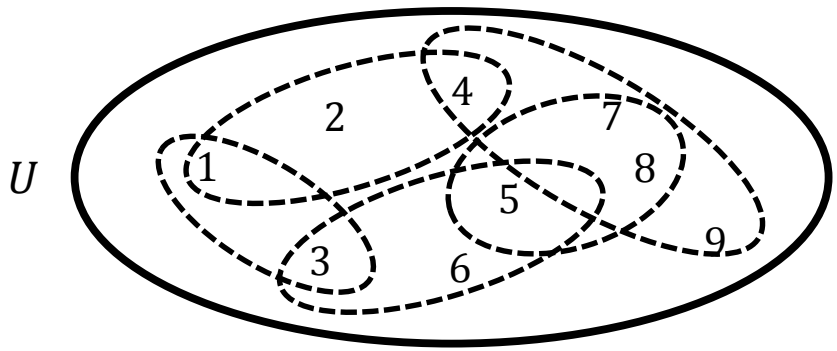

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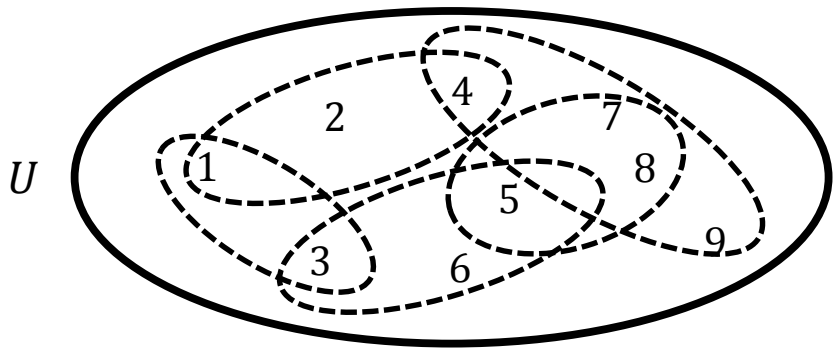

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SET COVER is **NP**-Complete

Theorem: VERTEX COVER $\leq_p$ SET COVER

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VERTEX COVER $\leq_p$ SET COVER

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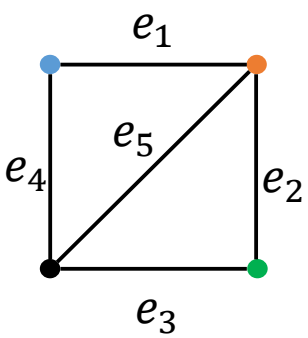

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$$U = \{e_1, e_2, e_3, e_4, e_5\}$$

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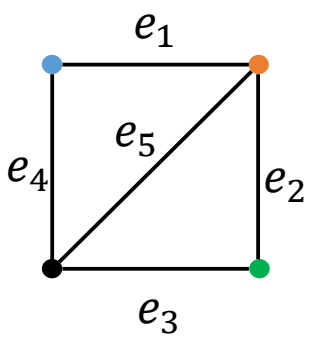

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Reduction:

- Let $\langle G = (V, E), k\rangle$ be an instance of VERTEX COVER
- Set $U := E$
- For every vertex $v \in V$ create a set S_v

$$S_v := \{e \in E \mid e \text{ is incident to } v\}$$

- Let $\mathcal{F} := \{S_v \mid v \in V\}$ and set $k' = k$

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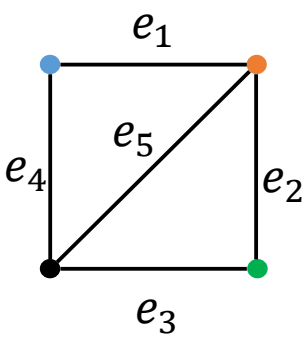

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Efficiency: Obvious from construction

Correctness ($\Rightarrow$): If G has a vertex cover of cardinality k, then U can be covered by k sets

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- Suppose $C \subseteq V$ is a vertex cover of G and $|C| = k$
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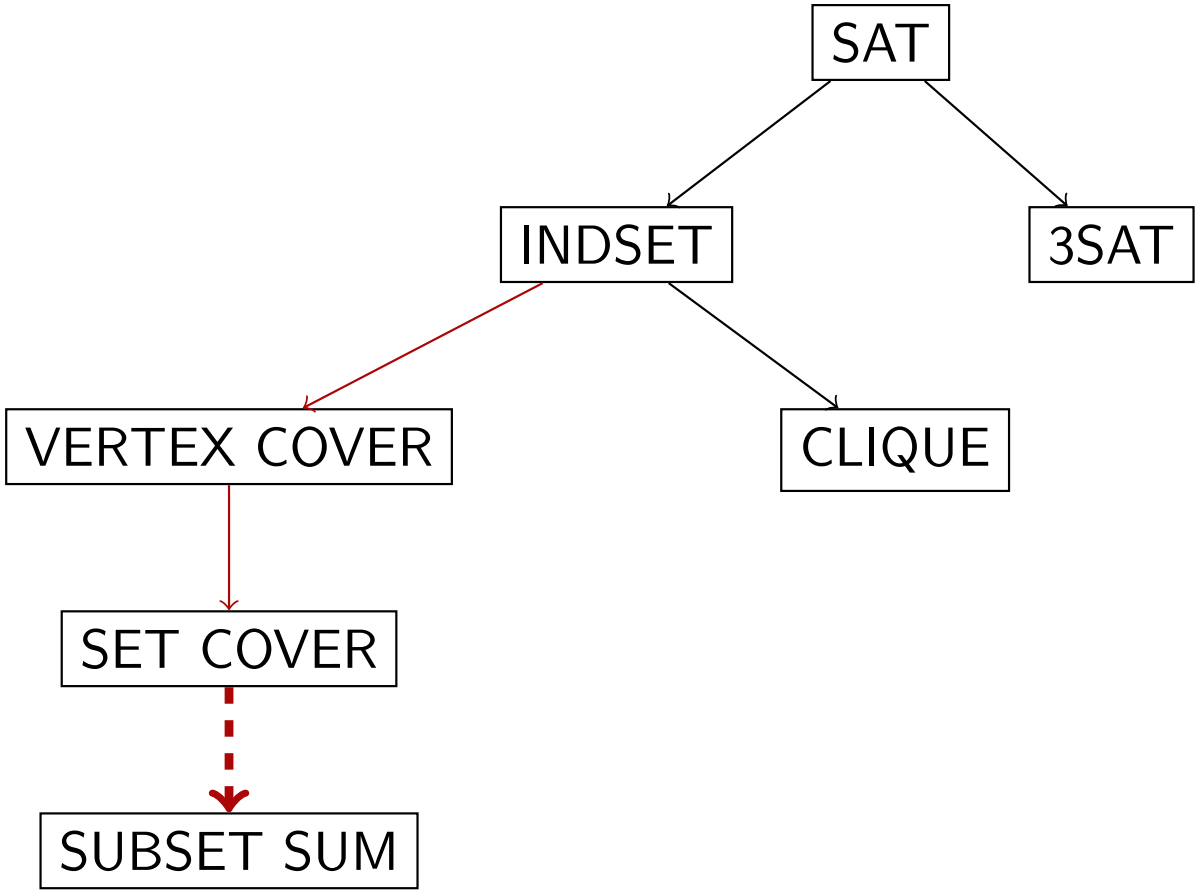
Proof:

- Let $S_{v_1}, S_{v_2}, \ldots, S_{v_k}$ be a collection of sets which cover $U = E$
- We claim that $C = \{v_1, v_2, \ldots, v_k\}$ is a vertex cover of G
- Indeed, every edge e in G belongs to S_{v_i} for some $i \in \{1, 2, \ldots, k\}$
- Hence, every edge e in G is incident to some vertex $v_i \in C$


SAT
INDSET
3SAT
VERTEX COVER
CLIQUE
SET COVER
SUBSET SUM

SUBSET-SUM

Let X denote a (multi) set of positive integers

Definition: SUBSET-SUM
$= \{\langle X, s\rangle : X$ contains a subset whose elements sum to $s\}$

Example: $X = \{1, 3, 4, 6, 13, 13\}$

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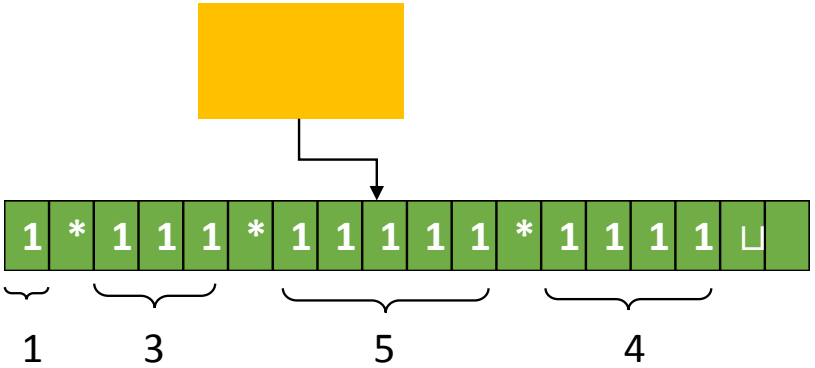
It depends on the input length

In $SUBSET\text{-}SUM$ we use binary encoding

Unary vs Binary



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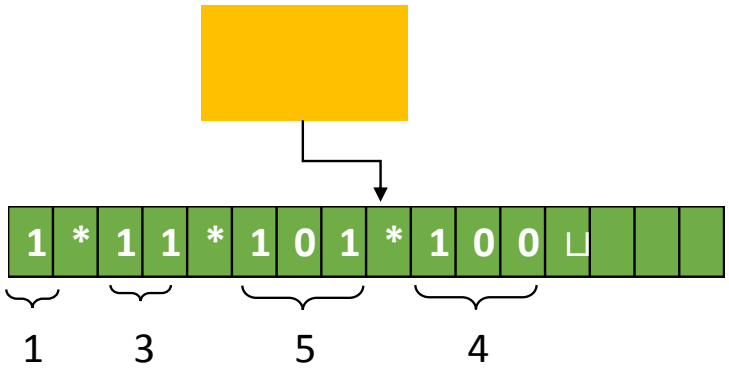

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Input length: $x_1 + x_2 + \cdots + x_m + s$ $\gg$ Input length: $\lceil\log(x_1)\rceil + \cdots + \lceil\log(x_m)\rceil + \lceil\log(s)\rceil$

SUBSET-SUM: An Algorithm

- Let $X = \{x_1, x_2, \dots, x_m\}$, $\quad sum \coloneqq \sum_{j=1}^{m} x_j$
- FOR $i = 1 \,..\, m$ DO
- FOR $j = 0 \,..\, sum$ DO
- $A[i,j] := False$
- FOR $i = 0$ to m DO
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- IF $A[i-1, j-x_i] = True$ OR $A[i-1,j] = True$ THEN $A[i,j] := True$
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Running time: For the input $\langle \{x_1, \dots, x_m\}, s \rangle$, the above algorithm takes $O\left(m \cdot \left(\sum_{i=1}^{m} x_i\right)\right)$ steps to output whether or not $\langle X, s \rangle \in SUBSET\text{-}SUM$

Unary representation: The length of the input $x_1 + x_2 + \cdots + x_m + s$. In this case the running time is **polynomial** in the input length

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SUBSET-SUM is **NP**-Complete

Theorem: SET COVER $\leq_p$ SUBSET-SUM

Proof: Next lecture

Corollary: SUBSET-SUM is **NP**-Complete